

Midterm Exam Solutions

Statistical Methods(통계적방법론)

2026 1st semester

Section(교반): A1 Cadet Number(교번): _____ Name(성명): _____ Score(점수): _____

- All solutions must include a detailed step-by-step explanation.

1. [50 points] Consider the multiple linear regression model:

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \eta^{-1}) \text{ independently for } i = 1, \dots, n, \quad (1)$$

where $\mathbf{x}_1, \dots, \mathbf{x}_n, \boldsymbol{\beta} \in \mathbb{R}^p$ and $X = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top \in \mathbb{R}^{n \times p}$ is full rank. ($\text{rank}(X) = p$ with $n > p$) Letting $\mathbf{y} = (y_1, \dots, y_n)^\top$ and $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^\top$, (1) can equivalently be written as

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim MVN(\mathbf{0}, \eta^{-1}I_n).$$

The log-likelihood of $\boldsymbol{\beta}$ and η is

$$\ell(\boldsymbol{\beta}, \eta) = -\frac{n}{2} \log(2\pi) + \frac{n}{2} \log \eta - \frac{\eta}{2} (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

(1) Show that the score function of $\boldsymbol{\beta}$ and η is

$$S(\boldsymbol{\beta}, \eta) = \begin{pmatrix} \frac{\partial \ell}{\partial \boldsymbol{\beta}} \\ \frac{\partial \ell}{\partial \eta} \end{pmatrix} = \begin{pmatrix} \eta X^\top (\mathbf{y} - X\boldsymbol{\beta}) \\ \frac{n}{2\eta} - \frac{1}{2} (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}) \end{pmatrix}.$$

(2) Show that the maximum likelihood estimator(MLE) that solves $S(\boldsymbol{\beta}, \eta) = 0$ is

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}, \quad \hat{\eta} = \frac{n}{(\mathbf{y} - X\hat{\boldsymbol{\beta}})^\top (\mathbf{y} - X\hat{\boldsymbol{\beta}})}.$$

(Hint: find $\hat{\boldsymbol{\beta}}$ first.)

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(3) Show that the Fisher information matrix is

$$\mathcal{I}(\beta, \eta) = -E \begin{bmatrix} \frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} & \frac{\partial^2 \ell}{\partial \beta \partial \eta} \\ \frac{\partial^2 \ell}{\partial \eta \partial \beta^\top} & \frac{\partial^2 \ell}{\partial \eta^2} \end{bmatrix} = \begin{bmatrix} \eta X^\top X & \mathbf{0} \\ \mathbf{0}^\top & \frac{n}{2\eta^2} \end{bmatrix}.$$

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(4) Using the Fisher information in problem (3), find the asymptotic distribution of $\hat{\beta}$.

(5) Consider the hypothesis test:

$$H_0 : \beta = \mathbf{0}, \quad H_1 : \beta \neq \mathbf{0}.$$

Show that the Wald test statistic is

$$W = (\hat{\beta}^\top \quad 0) \mathcal{I}(\hat{\beta}, \hat{\eta}) \begin{pmatrix} \hat{\beta} \\ 0 \end{pmatrix} = \frac{(X\hat{\beta})^\top (X\hat{\beta})}{(\mathbf{y} - X\hat{\beta})^\top (\mathbf{y} - X\hat{\beta})/n}.$$

What is the (asymptotic) null distribution of the test statistic?

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2. [30 points] Consider the logistic regression model with a single binary covariate $x_i \in \{0, 1\}$:

$$\log \frac{p_i}{1 - p_i} = \beta_0 + \beta_1 x_i, \quad y_i \sim \text{Bernoulli}(p_i) \text{ independently, } i = 1, \dots, n.$$

The data are summarized in the following contingency table:

		Y		
		0	1	Total
X	0	n_{00}	n_{01}	$n_{00} + n_{01} = n_0$
	1	n_{10}	n_{11}	$n_{10} + n_{11} = n_1$
	Total	$n_{00} + n_{10}$	$n_{01} + n_{11}$	$n_{00} + n_{01} + n_{10} + n_{11}$

where $n_{00}, n_{01}, n_{10}, n_{11} > 0$.

(1) Show that

$$p_i = \frac{1}{1 + \exp(-\beta_0 - \beta_1 x_i)}.$$

(2) For subjects with $x_i = 0$, $p_i = p(0) = \frac{e^{\beta_0}}{1 + e^{\beta_0}}$; for $x_i = 1$, $p_i = p(1) = \frac{e^{\beta_0 + \beta_1}}{1 + e^{\beta_0 + \beta_1}}$. Express the odds ratio (OR)

$$\text{OR} = \frac{p(1)/(1 - p(1))}{p(0)/(1 - p(0))}$$

in terms of β_0 and β_1 .

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(3) The MLEs of $p(0)$ and $p(1)$ are given as

$$\hat{p}(0) = \frac{n_{01}}{n_{00} + n_{01}}, \quad \hat{p}(1) = \frac{n_{11}}{n_{10} + n_{11}}.$$

Show that the maximum likelihood estimator (MLE) of log odds ratio (OR) is

$$\log \widehat{\text{OR}} = \log \frac{\hat{p}(1)/(1 - \hat{p}(1))}{\hat{p}(0)/(1 - \hat{p}(0))} = \log \frac{n_{11} n_{00}}{n_{10} n_{01}}.$$

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3. [20 points] Consider the multiple linear regression model in problem 1.

(1) Find the ridge estimator of β , denoted $\hat{\beta}_\lambda$, that minimizes

$$\ell(\beta) = (\mathbf{y} - X\beta)^\top (\mathbf{y} - X\beta) + \lambda \beta^\top \beta = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 + \lambda \beta^\top \beta.$$

where $\lambda \geq 0$.

(2) Find $E(\hat{\beta}_\lambda)$. What is the value of λ that makes $E(\hat{\beta}_\lambda) = \beta$?